INCH-POUND
MIL-DTL-24643/65A
28 January 2010
SUPERSEDING
MIL-DTL-24643/65
1 March 2007

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3C179DT (INCLUDING VARIATION LS6C179DT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS3C179DT (three internal coaxial cables) and LS6C179DT (six internal coaxial cables).

Construction (Non-Watertight)

First	-	Copper conductor, coated or uncoated, in accordance with ASTM B286, 0.0122-inch diameter.
Second	-	Thermoset insulation, diameter 0.056 inch nominal.
Third	-	Light bonded aluminum foil-polyester tape, aluminum foil shield. Foil in contact with braid.
Fourth	-	Braided shield of 38 AWG tin-coated copper: minimum coverage 95 percent, diameter 0.079 inch nominal.
Fifth	-	Cross-linked polyolefin jacket. Diameter 0.100±0.005 inch.
Sixth	-	The required number of coaxial cables cabled together with a lay in accordance with 3.4.9 of MIL-DTL-24643. Fillers may be used as necessary to make a firm, well-rounded assembly.
Seventh	-	An optional binder.
Eighth	-	Tinned copper braid, 36 gauge, 90 percent coverage.
Ninth	-	Cross-linked polyolefin jacket. Minimum thickness of 0.040 inch.

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TABLE I. Details.

Military part no. M24643/65	Size	Number of coaxes	Diameter over shield (inches nominal)	Cable jacket thickness min. avg. (inches)	Overall diameter (inches)		Cold bend mandrel diameter (inches)
					Min.	Max.	
-01UO	3	3	0.240	0.047	0.335	0.365	4.0
-02UO	6	6	0.325	0.050	0.425	0.455	5.0

REQUIREMENTS:

Qualification Required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	110
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield	1000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Jacket flaws	No failure
Conductor and shield continuity	No failure

Group A:

Visual and dimensional	No failure
Characteristic impedance (ohms) (TDR method in accordance with characteristic impedance of MIL-DTL-17)	75±5

Group B:

Thermoset proof test (percent, max.)	
Overall jacket	50
Cold bending, cable mandrel diameter 12 times cable diameter	No damage
Bending endurance (at 25+2 °C, cycles, min.)	25
Tear (lb/in thickness, min.) (ASTM D470)	35
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	600

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Elongation (percent, min.)	200
Jacket (component)	
Tensile strength (lb/in^2, min.)	1000
Elongation (percent, min.)	100
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Group C:	
Physicals (aged) (air oven)	
Insulation	
Tensile strength (percent of unaged, min.)	80
Elongation (percent of unaged, min.)	80
Jacket (components)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	125
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging stability (cable) (98±2 °C) (Test in accordance with aging stability of MIL-DTL-17)	No failure
Dimensional stability (cable) (85±2 °C) (Test in accordance with dimensional stability of MIL-DTL-17)	No failure
Abrasion resistance (overall jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket (component and overall)	2
Fillers	2
Insulation	18

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Shield insulation	2
Halogen content (percent, max.)	
Jacket (component and overall)	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket (component and overall)	25
Fillers	45
Insulation	45
Toxicity index, max.	
Jacket (component and overall)	5
Fillers	5
Insulation	1.5
Durometer (overall jacket) – Type A (hardness, min.)	80
Weathering (overall jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS3C179DT, LS6C179DT	All sizes 500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-108)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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